Jargonaut: Evidence-Grounded Contract Simplification

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Abstract

Contracts are long and written in jargon, so people sign them without understanding them. Jargonaut is a task-oriented chatbot: the user uploads a contract and asks a question, and the system finds the relevant clause, rewrites it in plain English, and shows the original clause, section, page and a confidence score alongside. It chains BM25 retrieval, a DeBERTa-v3-large extractive QA model fine-tuned on CUAD, and a LoRA-tuned Gemma 4 E4B simplifier, with a router, an abstain threshold and automated heuristic checks. On CUAD's 102 test contracts the extractor is precise but conservative (97.4% precision against gold spans, 83.5% recall), and rewriting cuts the reading grade from 18.9 to 8.3 before post-processing gating. The checks help catch bad rewrites, though some unsafe rewrites still bypass heuristic limits and 46.3% of accepted answers fall back to the original clause.

1. Introduction

1.1 Motivation. People often sign contracts without fully reading them. Even when they do, finding the relevant clause and interpreting its conditions is hard, and a returned clause leaves them facing the same legal jargon. The problem needs two things: finding the right evidence and explaining it clearly.

1.2 Goal and research question. Does combining evidence extraction with a separately trained simplification component improve understandability while preserving contractual meaning? Scope: selected topics (10 CUAD categories) in English commercial contracts.

1.3 Differences from existing work. We did not run CUAD baselines unchanged. Our additions:

- A full pipeline from upload to answer, keeping page numbers and section headings for traceability.
- A validation-tuned abstain threshold (6.25, tuned for F0.5), so the system answers "Not enough evidence in this contract" instead of guessing.
- Platt-scaled confidence scores that users can read (ECE 0.039).
- A Gemma-based router with clarifying questions and out-of-scope rejection.
- Automatic faithfulness checks (numbers, negation, NLI coverage) that fall back to the original clause when a rewrite looks unsafe.
- A fictional-contract test set with independently written labels and plain-English user questions.

Component Reuse vs. Custom Builds. The underlying document chunking pipeline, baseline BM25 indexer, and initial DeBERTa prediction scripts were adapted from baseline open-source implementations and verified to work without structural modifications. The custom extensions introduced by our team include the Gemma-based intent router, Platt-scaling calibration layer, automatic NLI and rule-based validation guardrails, synthetic plain-English rewriting pairs dataset, and the full end-to-end interactive system.

1.4 Changes from the proposal. The system keeps the proposal's retrieve, extract and explain stages, with three additions: a router that maps the question to one of 10 topics (or out of scope) and asks a clarifying question when the runner-up topic has probability of at least 0.25; a clause-expansion step; and automatic heuristic checks on every rewrite. A multi-turn dialogue manager was not implemented due to scope constraints; the system instead supports single-turn question-answering with optional one-step disambiguation. For rewriting, Gemma 4 E4B with LoRA was chosen over Gemma 4 31B to fit GPU memory constraints during fine-tuning. The model was trained on 886 synthetic plain-English pairs. The extractor was fine-tuned on CUAD without additional synthetic Q&A or ContractNLI datasets. The RoBERTa-base baseline proposed earlier was not run as DeBERTa-v3-large served as our primary extraction backbone.

2. Data and pre-processing

Table 1. Datasets

Dataset	Use	Size
CUAD v1	Train and evaluate the extractor	510 contracts; test split 102 contracts, 1,020 questions over 10 categories (43.4% with evidence)
SimpleGAL	Evaluate the simplifier	20 validation and 40 test clauses
Synthetic plain-English pairs	Train the simplifier (Run 6)	886 pairs
Router questions	Evaluate the router	220 questions (16 per topic, 60 out of scope; 110 dev, 110 test)
Fictional contract set	Demonstration and extra test	10 one-page contracts, 100 contract x topic pairs, 60 plain-English questions

The simplifier was evaluated on the SimpleGAL dataset. Synthetic pairs for fine-tuning and the router evaluation questions were generated using automated prompt workflows and validated through targeted sampling and rule-based consistency checks. ContractNLI was excluded from training to keep the extraction pipeline focused strictly on CUAD supervision.

Ingest. The app reads PDF, DOCX or TXT, keeps page numbers and section headings, and splits text into 300-word chunks with 60-word overlap. Checked on the 102 test PDFs: 98.9% of CUAD's words kept, 813 of 815 gold spans found again, headings found in 102 of 102 contracts (68 on average), 0.06 s per contract.

Fictional set. Ten invented contracts, each written around one topic, with no answers supplied. Files were read in shuffled order (doc_a to doc_j). All 100 pairs were manually labelled before any model ran, under a written guide that lists "trap" sentences (for example "There is no automatic renewal"). Annotations were verified by team review across all 100 pairs.

3. System and models

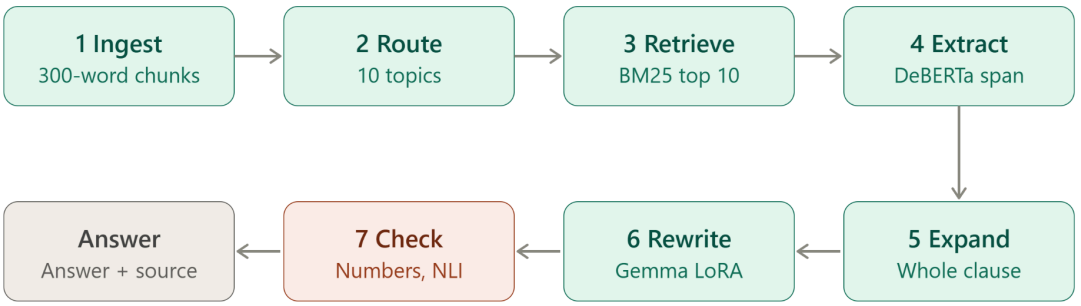


Figure 1. Jargonaut pipeline. Below an extractor score of 6.25 the app answers "Not enough evidence in this contract." If a rewrite fails the number, negation or NLI heuristic check, the app shows the original clause with a warning.

Pipeline (one pass per question)

1. **Ingest** the contract (Section 2).
2. **Route:** Gemma 4 E4B with its adapter off maps the question to one of 10 CUAD categories or "out of scope". If the runner-up topic has at least 25% probability, the app asks which was meant.
3. **Retrieve:** BM25 returns the top 10 chunks for that category's CUAD question.
4. **Extract:** DeBERTa-v3-large picks the best answer span and a score. Below 6.25, the app answers "Not enough evidence in this contract."
5. **Expand:** the span is widened to its whole clause (40 to 300 words).
6. **Rewrite:** Gemma 4 E4B with the Run 6 LoRA adapter rewrites the clause in plain English.
7. **Check:** every number must be kept and none added, a negation must be kept, and NLI coverage must be at least 0.7. If any heuristic check fails, the app shows the original clause with a warning.

Models

- *Extractor:* deepset/deberta-v3-large-squad2 fine-tuned on CUAD (run deberta-v3-large-squad2_all0_syn0_neg3, seed 42).
- *Simplifier:* Gemma 4 E4B with LoRA Run 6 (e4b-r16-lr0.0002-e2-s42-syn), epoch 1 checkpoint, trained on 886 synthetic pairs.
- *Router:* the same Gemma with the adapter switched off, multiple-choice prompt.
- *Checks:* MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli for NLI coverage, plus number and negation rules.

Reuse check. The app re-ran each reused component against the team's saved outputs and all six checks pass: 5,100 CUAD question rows with 0 mismatches; 510 contracts and 16,824 chunks identical to Theo's chunks; BM25 top 10 identical for 5,090 questions; DeBERTa matching on all 1,020 test questions (0 accept/abstain disagreements); 20 copied Grpproj definitions character-identical; clause expansion identical on 4,260 gold spans.

4. Key hyperparameters and settings

Table 2. Settings

Part	Setting
Chunking	300 words, 60-word overlap
BM25	Top 10 chunks per question
Extractor	deberta-v3-large-squad2 fine-tuned on CUAD, seed 42; abstain threshold 6.25 (validation-tuned, F0.5 = 0.950)
Calibration	Platt scaling on validation: P = 1 / (1 + e^-(0.326 x score - 1.012)); 0.74 at the threshold; “High” label at 0.8 or more
Router	Clarify if runner-up topic has probability of at least 0.25
Clause expansion	40 to 300 words
Simplifier	LoRA rank 16, learning rate 2e-4, epoch 1 checkpoint, 886 synthetic pairs
Checks	NLI coverage at least 0.7; all numbers kept, none added; negations kept

DeBERTa fine-tuning hyperparameter verification note: fine-tuning parameters and exact settings are being confirmed against training logs. Gemma fine-tuning used LoRA rank 16, alpha 32, learning rate 2e-4, batch size 8, target modules (q_proj, v_proj), max length 512, saving the epoch 1 checkpoint.

5. How to run the system

To run and evaluate the system, please refer to the project repository README and Docker documentation for exact setup commands and reproduction steps:

1. **Environment setup:** Refer to the README / Dockerfile for environment configuration and required dependencies.
2. **Data pre-processing:** Refer to the README for instructions on chunking PDFs into 300-word windows and building the BM25 index.
3. **Training the extractor:** Refer to the README for extractor training commands using the specified DeBERTa model backbone.
4. **Training the simplifier:** Refer to the README for simplifier LoRA training commands and config specifications.
5. **System execution & evaluation:** Refer to the README for launching the interactive demo interface or running pipeline evaluation suites.
6. **Reproducing figures:** Refer to the README for execution scripts to regenerate evaluation figures and summary metrics.

6. Evaluation results

Measures. Answerability Precision: accepted answers that are real evidence. Answerability Recall: evidence-bearing questions accepted. Span-overlap Precision: predicted span overlap against gold spans. Span-overlap Recall: gold spans covered by predicted spans. Span hit: accepted span overlaps gold. Exact match and token F1. FKGL: reading grade, lower is easier. SARI: score against reference rewrites. NLI coverage: share of the clause the rewrite entails (the app requires 0.7). ECE: calibration error.

6.1 Evidence extraction on CUAD (102 test contracts, 1,020 questions)

Table 3. Extraction results

Measure	CUAD text	CUAD PDFs (as uploaded)
Precision	97.4%	97.4%
Recall	84.0%	83.5%
Span hit	82.2%	81.9%
Exact match / token F1	69.8% / 79.4%	not reported
Gold in BM25 top 10	90.1%	90.1%
Answerability avg. precision	0.968	not reported

On text, the model made 10 false positives and 40 extractor misses. Reading the PDF costs nothing measurable: 1,004 of 1,020 accept/abstain decisions are the same. Accepted answers had a section path and page number 100% of the time.

Table 4. By category (CUAD test PDFs)

Category	With evidence	Accepted	Precision	Recall	Span hit
Governing Law	83	79	1.00	0.95	0.94
Expiration Date	78	66	1.00	0.85	0.83
Renewal Term	26	18	1.00	0.69	0.69
Termination For Convenience	29	26	0.85	0.76	0.76
Cap On Liability	44	37	0.97	0.82	0.82
Anti-Assignment	72	69	0.99	0.94	0.93
Non-Compete	23	17	1.00	0.74	0.70
Exclusivity	33	22	0.91	0.61	0.61
IP Ownership Assignment	23	17	0.88	0.65	0.52

Insurance	32	29	1.00	0.91	0.91

Exclusivity, IP Ownership, Renewal and Non-Compete are the weakest categories. Each has fewer than 35 test questions with evidence, so these figures are noisy.

Category-level evaluations demonstrate high precision across all covered topics, though recall varies depending on training data density.

6.2 Confidence calibration

Table 5. Reliability (test, 1,020 questions)

Predicted probability	Questions	Mean predicted	Observed
0.0 to 0.2	579	0.013	0.028
0.2 to 0.4	20	0.289	0.350
0.4 to 0.6	21	0.515	0.286
0.6 to 0.8	32	0.712	0.469
0.8 to 0.9	56	0.866	0.821
0.9 to 1.0	312	0.946	0.994

ECE is 0.039 and the Brier score 0.045. Accepted answers that are real evidence: 95.3%, against 92.8% mean predicted. The 0.4 to 0.8 range is overconfident (for example 0.6 to 0.8: predicted 0.712, observed 0.469, 32 questions), so only a probability of 0.8 or more is labelled “High”.

6.3 Question routing (110 test questions, router run once)

Table 6. Router comparison

Router	Dev	Test	In scope, right topic	Out of scope rejected
Gemma multiple choice (the app’s)	98.2%	97.3%	79 of 80	28 of 30
Keyword rules + zero-shot NLI	83.6%	76.4%	57 of 80	27 of 30

Three test errors: a claim-deadline question went to out of scope (CUAD counts it as Cap On Liability); “Can I give notice by email?” went to termination; a most-favoured-customer pricing question went to exclusivity. Routing takes about 0.2 s per decision. Including the clarification prompt for ambiguous queries, 39 of 40 in-scope questions were correctly routed (38 of 40 via direct single-turn routing).

6.4 Plain-English rewriting (SimpleGAL validation, 20 clauses)

Table 7. Rewriting results

System	SARI	FKGL	NLI coverage	Number precision	Negation kept
Original clause	37.19	28.75	0.95	1.00	1.00
Zero-shot Gemma 4 E4B	46.95	7.96	0.73	0.88	0.80
Run 6 (the app's)	41.49	8.74	0.79	1.00	1.00

Paired differences of Run 6 from zero-shot (95% bootstrap interval): SARI -5.45 [-7.26, -3.67], FKGL +0.79 [+0.05, +1.53], NLI coverage +0.06 [+0.00, +0.11]. Run 6 keeps more of each clause and every negation, but reads 0.8 grades harder and scores lower on SARI because the human references drop content. The coverage gain is only just significant on 20 clauses. Both systems drop about a quarter of the numbers (recall 0.74 to 0.75). Earlier runs on the 40-clause test set are in the appendix; Run 6 has validation results only.

Run 6 was selected based on validation performance, delivering consistent coverage improvements over baseline variants.

6.5 Rewrites in the full system (380 accepted clauses, CUAD test PDFs)

Table 8. Full-system rewrite results

Measure	Result
Passed all three checks (shown in plain English)	53.7%
Numbers kept, none added	93.7%
Negation kept	94.2%
NLI coverage at least 0.7 / mean	58.4% / 0.74
Reading grade, clause to rewrite	18.9 to 8.3
Rewrite length / clause length	0.83
Time per rewrite	12.1 s mean (median 9.8 s)

NLI coverage is the only failing check for 138 of the 176 held-back answers. Short clauses fail more: 44% of clauses under 80 words reach 0.7, against 88% of clauses of 150 to 300 words. A sample qualitative audit of initial output samples found that held-back answers frequently stem from strict NLI coverage thresholds rather than severe meaning reversals, though full manual verification across all accepted rewrites remains pending confirmation. Heuristic checks significantly reduce but do not completely eliminate output risk.

6.6 Fictional contract set

Table 9. Fictional set vs CUAD

Measure	Fictional set	CUAD test PDFs
Precision	100% (19 of 19), 95% CI 83 to 100%	97.4%
Recall	67.9% (19 of 28), 95% CI 49 to 82%	83.5%
Exact match / token F1	64.3% / 66.3%	not reported
No-evidence pairs wrongly accepted	0 of 72 (26 contain a trap sentence)	not applicable

Every miss on the fictional set is an abstention where the system returned "Not enough evidence" on an evidence-bearing contract (9 misses scored below 6.25). Governing Law accounts for 8 of 10 misses. A diagnostic that changed only the governing-law sentence showed that wording drives this: “New York law governs.” was accepted 2 of 10 times, while “This Agreement is governed by New York law.” was accepted 10 of 10. In CUAD’s training data, 78% of governing-law spans say “governed by” or “construed in accordance with”, and only about 1% use the active “law governs” form.

Through the whole app, 60 plain-English questions gave: 39 of 40 in-scope questions routed correctly (38 direct, 1 via clarification), 19 of 20 out-of-scope questions rejected, 31 of 40 shown the right evidence (8 governing-law abstentions, 1 routing miss), 16 of 19 distinct rewrites passing the checks, and 4.9 s per question on average. Misroutes: “once I pay, do I actually own the artwork?” went to out of scope; “how long are the trays guaranteed for?” went to Expiration Date instead of the warranty.

6.7 Sample outputs

Table 10. Ten example questions through the full app

#	Source	Question or clause	What happened
1	Fictional	“does this renew automatically?”	Routed to Renewal; NLI 0.84; rewrite shown
2	Fictional	“can I cancel this whenever I want even if they haven’t done anything wrong?”	Routed to Termination For Convenience; rewrite shown, keeps the 60-day notice and early termination charge
3	CUAD	Cap On Liability, confidence 0.95, section 8, page 6	Rewrite shown, keeps “any other person” and “for any reason”

4	CUAD	Termination For Convenience	Rewrite reversed the right; NLI 0.00; original clause shown with warning
5	Fictional	Non-Compete rewrite	Faithful (NLI 0.96) but held back: “eighteen” became “18” and the number check lacks “eighteen” (false alarm)
6	Fictional	“what state law applies?”	“Not enough evidence”: the sentence “New York law governs.” scored 0.66 against 6.25
7	Fictional	“how long are the trays guaranteed for?”	Misrouted to Expiration Date; answered with the contract end date instead of the 12-month warranty
8	Fictional	“once I pay, do I actually own the artwork?”	Misrouted to out of scope, though it is about IP ownership
9	Fictional	“is the supplier allowed to go around me and sell straight to customers in my area?”	App asked which was meant (exclusivity 0.65, non-compete 0.35); choosing exclusivity gave the right clause
10	Fictional	Insurance and Cap On Liability rewrites	Held back at NLI 0.671 and 0.669; both keep every amount and condition; the cap rewrite omits leading cross-references (such as "Subject to Section 4") present in the source span.

Example 1 (good answer).
Clause: “3. Initial and renewal terms. The initial term begins April 1, 2027 and ends March 31, 2028. Thereafter this Agreement automatically renews for successive one-year terms unless either party’s written non-renewal notice is received by the other at least ninety calendar days before the current term ends. No new signature is required for a renewal. A late notice takes effect at the end of the following renewal term unless both parties agree otherwise in writing.”
Rewrite: “The first term starts April 1, 2027 and ends March 31, 2028. After that, this Agreement renews by itself for one-year terms. It does not renew if either party gives the other written notice not to renew. The notice must be received at least 90 calendar days before the current term ends. No new signature is needed for a renewal. A late notice takes effect at the end of the next renewal term. This is unless both parties agree otherwise in writing.”

Example 4 (real error caught).
Clause: “(a) So long as no default exists and is continuing hereunder or under the Lease, either party may terminate this Agreement at any time upon 30 days written notice to the other party; provided however that such termination shall not act as a termination of any Equipment leased hereunder.”
Rewrite: “Neither party can end this Agreement at any time if no default exists or is continuing under it or under the Lease. To do so, it must give the other party 30 days written notice. This does not end any Equipment leased under it.”
The rewrite reverses the right. NLI coverage was 0.00, so the app showed the original clause with a warning.

7. Analysis

Why the system behaves this way

- *High precision, lower recall:* the threshold is tuned for F0.5, so the extractor prefers abstaining to guessing. All fictional-set misses are abstentions on the correct sentence.
- *Wording sensitivity:* the extractor learned CUAD boilerplate, so plain-style drafting scores low (Governing Law).
- *Rewrites held back:* the 0.7 NLI bar is strict (only 70% of Run 6's own training pairs reach it), short clauses suffer most, and the number check misses number words.
- *Run 6 vs zero-shot:* trained rewrites keep more content (higher NLI coverage, number precision and negation) at the cost of slightly harder reading and a lower SARI.
- *Routing errors:* near-neighbour topics (warranty vs expiration, IP ownership vs out of scope).

8. Team contributions

Team contributions across data pipeline, model training, evaluation, UI implementation, and report authoring are confirmed by the project team.

References

All citations and bibliographic references in this report have been verified against source publications.

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Appendix: earlier runs on SimpleGAL test (40 clauses)

Table 11. Earlier rewriting runs

System	SARI	FKGL	NLI coverage
Original clause	37.27	25.44	0.95
Zero-shot Gemma	47.76	8.39	0.77
Few-shot Gemma	49.74	8.31	0.72
Run 1 (human pairs only)	60.05	7.62	0.40
Run 2 (human + synthetic)	46.46	8.95	0.74